The Fundamental Theorem of Calculus, pt. 1

Evaluate the integral $\int_0^8 3dx$ using the FTC.

1. Find the antiderivative

Antiderivative of $f(x) = 3$ is $F(x) = 3x + C$

2. Apply FTC, pt. 1 : $\int_a^b f(x)dx = F(b) - F(a)$

$F(8) - F(0)$

$= (8x + C) - C$

$\cdot\, x = 3 \rightarrow (8(3) + C) - C = 24$

Evaluate $\int_0^3 (12x^5 + 12x^2 - 6x)dx$ using the FTC, pt. 1.

$f(x) = 12x^5 \rightarrow \frac{12x^6}{6} + C = 2x^6 = F(x)$

$f(x) = 12x^2 \rightarrow F(x) = \frac{12x^3}{3} = 4x^3$

$f(x) = -6x \rightarrow F(x) = \frac{-6x^2}{2} = -3x^2$

$F(3) - F(0)$

$(2(3)^6 + 4x^3 - 3x^2 + C) - (2(0)^6 + 4(0)^3 - 3(0)^2 + C)$

$= (1539 + C) - (C) = 1539$

Evaluate $\int_{-2}^2 (10x^9 + 25x^{19})dx$ using FTC, pt. 1

$f(x) = 10x^9 \rightarrow F(x) = \frac{10x^{10}}{10} = x^{10}$, $f(x) = 25x^{19} \rightarrow F(x) = \frac{25x^{20}}{20}$

$F(2) - F(-2)$

$\left((2)^{10} + \frac{25(2)^{20}}{20}\right) - \left((-2)^{10} + \frac{25(-2)^{20}}{20}\right)$

$1311744 - 1{,}311{,}744 = 0$

Evaluate $\int_0^4 5\sqrt{y}\,dy$ using FTC, pt. 1

$f(y) = 5\sqrt{y} \rightarrow F(x) = \frac{5y^{3/2}}{3/2} + C = \frac{10y^{3/2}}{3}$

or $5y^{\frac{1}{2}}$

$F(4) - F(0)$

$\left(\frac{10(4)^{3/2}}{3} + C\right) - \left(\frac{10(0)^{3/2}}{3} + C\right) = \frac{80}{3}$

$\int_1^4 77x^{5/2}\,dx$, FTC pt. 1

$f(x) = 77x^{5/2} \rightarrow F(x) = \frac{77x^{7/2}}{7/2} + C = 22x^{7/2} + C$

$F(4) - F(1) \rightarrow (22(4)^{7/2} + C) - (22(1)^{7/2} + C)$

$= 2794$

$\int_5^9 3/x^3\,dx$, FTC pt. 1

$f(x) = \frac{3}{x^3} \rightarrow F(x) = 3x^{-3} + C$

$\hookrightarrow F(x) = \frac{3x^{-2}}{-2} + C$

$F(9) - F(5)$

$\frac{3(9)^{-2}}{-2} - \left(\frac{3(5)^{-2}}{-2}\right) = -\frac{1}{54} + \frac{3}{50}$ (LCD=1350)

$= -\frac{25}{1350} + \frac{81}{1350} = \frac{28}{675}$

Use FTC pt. 1 to find the area of the region under the graph of function $f(x) = 2x^2$ on $[0, 5]$.

$\int_0^5 2x^2\,dx \qquad F(x) = \frac{2x^3}{3} + C$

$\frac{2(5)^3}{3} - \frac{2(0)^3}{3} = \frac{250}{3} - 0 = \frac{250}{3}$

FTC, pt. 1

If $g(x) = \int_a^x f(t)dt$, than $g'(x) = f(x)$

- aka, the derivative of the antiderivative gives the original function.
- $\frac{d}{dx}\left[\int_a^x f(t)dt\right] = f(x)$

★ F is the antiderivative of f.

1. $\frac{d}{dx}\left[\int_0^x \sqrt{t^2+4}\ dt\right] = f(x) = \sqrt{t^2+4}$

$f(t) = \sqrt{t^2+4}$

$\frac{d}{dx}\left[\int_0^x f(t)dt\right] = \frac{d}{dx}\left[F(t)\Big|_0^x\right]$

$\frac{d}{dx}\left[F(x) - F(0)\right] = f(x) - 0 = f(x)$

$f(x) = \sqrt{t^2+4}$

2. $\frac{d}{dx}\left[\int_x^4 \sqrt{t^3+5}\ dt\right] = -\sqrt{t^3+5}$

$f(t) = \sqrt{t^3+5}$

$\frac{d}{dx}\left[\int_x^4 f(t)\ dt\right] = \frac{d}{dx}\left[F(t)\Big|_x^4\right]$

$\frac{d}{dx}\left[F(4) - F(x)\right] = 0 - f(x)$

derivative of a constant is 0 $= -f(x)$

$f(x) = -\sqrt{t^3+5}$